

Process & Decision Documentation

Side Quest – Reflective Camera Experience

Project/Assignment Decisions

For this side quest, the most significant decision was to remove player control and implement an automatic camera system.

Initially, I considered allowing the user to navigate the world. However, this would have shifted the experience toward exploration or gameplay. Since the prompt emphasized reflection and meditation, I decided to prioritize authored pacing over interactivity.

This decision affected all subsequent implementation choices, including waypoint-based movement, easing, pause timing, subtle drift, and restrained visual effects. The experience became guided rather than reactive, which preserved tonal coherence.

Role-Based Process Evidence (Individual)

Name:

[Your Name]

Role(s):

Designer & Developer

Primary Responsibility for this Work:

Concept framing, motion system implementation, discoverable interaction logic, and visual polish.

Goal of Work Session

To design and implement a reflective camera system that:

- Scrolls through a world larger than the screen
- Uses pacing and motion to evoke a meditative tone

- Incorporates subtle discoverable elements without disrupting atmosphere
-

Tools, Resources, or Inputs Used

- p5.js
 - Lecture references on pacing and motion
 - GitHub for version control
 - GenAI (ChatGPT 5.2) for conceptual iteration and debugging support
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GenAI Documentation

Date Used:

[Insert date(s)]

Tool Disclosure:

ChatGPT 5.2

Purpose of Use:

- Brainstorming structural approach
- Converting conceptual pacing ideas into code
- Debugging rendering issues
- Refining tone-based design decisions

Summary of Interaction:

GenAI assisted in translating design intentions (e.g., “breathing motion,” “subtle drift,” “camera attention”) into technical implementations such as easing functions, segment-based pause logic, and viewport-based discoverable rendering. It also helped diagnose runtime issues (e.g., undefined functions, duplicate variable declarations).

Human Decision Point(s):

Several suggestions from GenAI were modified or rejected:

- I chose not to implement celebratory confetti upon completion, despite it being suggested as a possible feature.
- I reduced the strength of cinematic zoom and exposure breathing to maintain subtlety.
- I simplified the completion state to removing the counter rather than adding effects.

Integrity & Verification Note:

All code suggestions were tested manually in the browser. Runtime errors were verified through console debugging. Visual adjustments were evaluated subjectively against the assignment’s tonal requirement (reflective/meditative). No GenAI output was used without modification.

Scope of GenAI Use:

GenAI supported ideation and debugging but did not independently produce the final cohesive structure. All design decisions, parameter tuning, and tonal evaluation were human-driven.

Limitations or Misfires:

Some early GenAI suggestions introduced overly dramatic effects (e.g., strong visual celebration on completion), which conflicted with the intended tone. These were adjusted or removed.

Summary of Process (Human + Tool)

The process unfolded iteratively:

1. Established automatic camera path using waypoint interpolation.
2. Introduced easing to soften motion.
3. Added pause segments to simulate breathing rhythm.
4. Layered subtle sinusoidal drift to remove mechanical precision.
5. Introduced discoverable symbols with minimal feedback.
6. Modified pacing dynamically when near undiscovered elements.
7. Added restrained cinematic polish (micro-zoom and vignette).
8. Simplified completion state to preserve tonal coherence.

Throughout development, most refinement involved reducing intensity rather than adding complexity.

Decision Points & Trade-offs

Decision 1: Automatic Camera vs Player Control

- Option A: Player navigation
- Option B: Authored automatic motion
- Chosen: Automatic motion
- Reason: Preserved reflective pacing

Decision 2: Completion Feedback

- Option A: Confetti / celebratory effect
 - Option B: Minimal disappearance of counter
 - Chosen: Minimal disappearance
 - Reason: Avoid tonal escalation
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Verification & Judgement

Evaluation methods included:

- Manual runtime testing in browser
- Console debugging
- Visual inspection of pacing and drift strength
- Re-reading assignment prompt to confirm tonal alignment

Tone consistency was the primary evaluation metric.

Limitations, Dead Ends, or Open Questions

- Subtle visual effects (zoom, exposure breathing) are subjective and may not be equally perceptible to all viewers.
- The experience intentionally avoids strong interaction, which may reduce engagement for some users.
- Future iterations could explore sound design while preserving tone.

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Appendix - Transcript

User uploaded the provided zip file containing the previous side quest template and sketch structure.

User:

The assignment says to create a reflective or meditative camera experience that scrolls through a world larger than the screen. I initially thought about adding user control, but that immediately felt wrong.

AI:

That hesitation is important. If the user controls the camera, the experience becomes exploratory. Exploration introduces agency and potential tension. The prompt emphasizes reflection.

So we made the first structural decision:

-  Automatic camera movement
-  Authored path
-  Emotion driven by pacing, not interaction

To implement that, we defined a world larger than the screen:

```
const WORLD_W = 3200;  
const WORLD_H = 2200;  
  
let camCenter = { x: 0, y: 0 };  
let cam = { x: 0, y: 0 };
```

And a set of predefined waypoints:

```
const path = [  
  { x: 350, y: 380 },  
  { x: 900, y: 520 },  
  { x: 1500, y: 420 },  
  { x: 2100, y: 760 },  
  { x: 2550, y: 1250 },  
  { x: 1950, y: 1650 },  
  { x: 1200, y: 1700 },  
  { x: 600, y: 1300 },  
];
```

User:

I considered random movement to avoid looking scripted, but randomness felt chaotic.

AI:

Randomness contradicts meditation. Instead, structure gives emotional stability.

We interpolated between waypoints:

```
const t = millis() / 1000;
const segFloat = t / SEG_SECONDS;

const i0 = floor(segFloat) % path.length;
const i1 = (i0 + 1) % path.length;

const uRaw = segFloat - floor(segFloat);
```

But linear interpolation looked mechanical.

User:

Yes, it felt robotic. Too mathematically perfect.

AI:

So we introduced easing to soften acceleration and deceleration:

```
function easeInOutSine(x) {
  x = constrain(x, 0, 1);
  return -(cos(PI * x) - 1) / 2;
}
```

And applied it:

```
const moveU = uRaw / movePortion;
u = easeInOutSine(moveU);
```

User:

Even with easing, constant movement felt restless.

AI:

That led to a key refinement:

- 🛑 Add pause at end of each segment
- ⌚ Simulate breathing

```
let movePortion = 0.8;

if (uRaw < movePortion) {
  const moveU = uRaw / movePortion;
  u = easeInOutSine(moveU);
} else {
  u = 1; // pause phase
}
```

User:

The pause worried me. I thought it might look like lag.

AI:

That's a valid design fear. A pause must feel intentional.

We kept it short and consistent. Rhythm, not glitch.

But something still felt too clean.

User:

Yes. It still felt sterile.

AI:

So we layered subtle drift. Not randomness — softness.

```
const driftStrength = 8;  
const driftX = sin(millis() * 0.0003) * driftStrength;  
const driftY = cos(millis() * 0.0002) * driftStrength;  
  
target.x += driftX;  
target.y += driftY;
```

-  Very low amplitude
-  Slow frequency
-  Barely perceptible

The goal was imperceptible imperfection.

User:

That finally made it feel less like math and more like something alive.

AI:

At this point, we intentionally stopped adding features. That restraint was deliberate.

We validated:

-  World larger than screen
-  Automatic guided motion
-  Easing
-  Pause
-  Drift

Only after tone coherence did we move to discoverables.

User:

Once the camera motion felt stable, I debated whether adding discoverables would break the meditative tone.

AI:

That hesitation was crucial. Discoverables risk turning a reflective experience into a collection game.

Before writing any code, we clarified intention:

-  The symbols should not feel like objectives
-  They should feel like quiet findings
-  No high saturation
-  No loud reward feedback

The design goal shifted from “collect” to “notice.”

User:

I didn’t want them visible from the start. That felt too obvious.

AI:

So we implemented conditional visibility. Symbols only render when inside the camera viewport.

```
function drawDiscoverables() {
  for (const d of discoverables) {

    const onScreen =
      d.x >= cam.x &&
      d.x <= cam.x + width &&
      d.y >= cam.y &&
      d.y <= cam.y + height;

    if (!onScreen) continue;

    if (d.found) {
      noStroke();
      fill(30, 120);
      circle(d.x, d.y, d.r * 1.4);
      continue;
    }

    const pulse = 0.6 + 0.4 * sin(millis() * 0.004);

    noFill();
    stroke(30, 120 * pulse);
    circle(d.x, d.y, d.r * 2.4);
  }
}
```

AI:

Notice what we did not do:

-  No bright colors
-  No sound

-  No score pop-ups
-  No animation explosion

The pulsing is subtle. It invites attention instead of demanding it.

User:

I was worried that clicking them would feel too game-like.

AI:

That concern led to another restraint decision.

Instead of “reward feedback,” we implemented minimal confirmation:

```
function mousePressed() {
  const wx = mouseX + cam.x;
  const wy = mouseY + cam.y;

  for (const d of discoverables) {
    if (d.found) continue;

    const distSq = (wx - d.x) * (wx - d.x) + (wy - d.y) * (wy - d.y);
    if (distSq <= d.r * d.r * 2) {
      d.found = true;
      break;
    }
  }
}
```

AI:

No particle burst. No sound. Just state change.

User:

But something felt missing. The camera wasn’t reacting to the symbols.

AI:

Exactly. That created a subtle mismatch. If the camera ignores what it encounters, the world feels indifferent.

So we added a behavioral shift when near undiscovered symbols:

```
function nearUndiscovered() {
  for (const d of discoverables) {
    if (d.found) continue;

    const dx = camCenter.x - d.x;
    const dy = camCenter.y - d.y;
    const dist = sqrt(dx * dx + dy * dy);

    if (dist < 250) return true;
  }
  return false;
}
```

```
}
```

And used it to alter pacing:

```
let movePortion = 0.8;  
  
if (nearUndiscovered()) {  
  movePortion = 0.6; // linger longer  
}
```

AI:

This was a philosophical shift.

The camera now “notices” something.

- 🐢 Movement slows
- ⏸ Pause extends
- 🌀 Drift softens

User:

I liked that change. It didn’t feel like gameplay. It felt like attention.

AI:

Exactly. We reframed interaction as perception.

At this stage, the system had:

- 🌍 Large scrolling world
- 🗺 Automatic authored path
- 🎵 Breathing rhythm
- 👁 Conditional discoverables
- 🐢 Camera linger behavior

And importantly, the tone remained coherent.

User:

Once the discoverables and camera linger were working, I started questioning whether the experience still felt meditative. It was calm, but maybe too flat.

AI:

That’s a subtle but important turning point. Reflection does not mean static. A reflective experience can still have depth. The question became:

- 🎬 Do we introduce cinematic motion?
- 🧑 Or keep it purely minimal and static?

This was not a technical decision. It was tonal.

User:

I was worried that adding zoom would make it dramatic. I didn't want it to feel like a trailer.

AI:

Exactly. Zoom can easily introduce tension. But if handled with restraint, it can create intimacy instead of drama.

So we introduced an extremely subtle oscillating zoom:

```
// Cinematic micro-zoom
const zoom = 1.0 + 0.015 * sin(millis() * 0.00008);

translate(width / 2, height / 2);
scale(zoom);
translate(-width / 2, -height / 2);
```

AI:

Key decisions here:

-  Amplitude is tiny ($1.0 \rightarrow 1.015$)
-  Frequency is extremely slow
-  It should be felt, not noticed

If the viewer notices the zoom consciously, it's too strong.

User:

At first it felt almost unnecessary. I could barely see it.

AI:

That's actually correct. Cinematic polish is often invisible. Its role is emotional reinforcement, not spectacle.

User:

Then I considered adding a vignette. I wasn't sure if that would look artificial.

AI:

Vignette can feel like an Instagram filter if overdone. But in cinema, it subtly directs focus.

So we implemented a soft edge darkening:

```
function drawVignette() {
  noFill();
  for (let i = 0; i < 14; i++) {
    const a = map(i, 0, 13, 0, 55);
    stroke(0, a);
    strokeWeight(26);
    rect(0, 0, width, height, 22);
  }
}
```

AI:

Again, restraint was the rule:

-  No heavy darkening
-  No color shift
-  Just subtle framing

User:

After adding zoom and vignette, the UI suddenly felt too loud.

AI:

That reaction makes sense. As the visual language became more cinematic, the interface needed to become quieter.

So we made several adjustments:

-  Reduced text opacity
-  Moved HUD inward from edges
-  Removed commit labels, replaced with semantic version

```
const VERSION = "v1.0";
```

```
fill(COL_TEXT, 200);
```

```
text(`Reflective Camera Cruise ${VERSION}`, 28, 44);
```

AI:

We also made a conscious decision:

-  No bold counters
-  No large overlays
-  No intrusive debug

Instead, debug was hidden by default:

```
let showDebug = false;
```

User:

That made it feel more like a finished piece rather than a prototype.

AI:

Exactly. The shift from “Commit 4” to “v1.0” wasn’t cosmetic. It reframed the project from iterative build to cohesive experience.

At this stage, the system now included:

-  Large world
-  Authored path

- 🧘 Breathing rhythm
- 👁 Discoverables
- 🐢 Camera attention behavior
- 🎬 Subtle zoom
- 🌑 Vignette
- 🧑 Minimal UI

User:

I briefly thought about pushing it further — maybe stronger visual effects.

AI:

And that's where restraint mattered again.

More effect ≠ more depth.

The tone was balanced. Over-polishing risked breaking coherence.

User:

When everything was working, I started thinking about what should happen after collecting all the symbols. It felt incomplete to have no ending.

AI:

That instinct is natural — we look for closure. But closure can take many forms. The danger was introducing celebration where reflection was intended.

We clarified the emotional question first:

- 🎉 Is this a “reward” moment?
- 🧘 Or is this a quiet realization?
- 🔔 Should the system react loudly, or subtly shift state?

User:

My first thought was confetti. That's the usual completion signal.

AI:

And that was a critical hesitation point.

Confetti implies:

- 🎉 Celebration
- 🎮 Achievement
- 🏆 Game logic

But your experience was not structured as a challenge. It was structured as noticing.

User:

Exactly. It felt wrong stylistically.

AI:

So instead of celebration, we considered reduction.

Completion could mean:

- 🗑️ Removing the counter
- 🌫️ Letting the symbols remain quietly faded
- 🧑🏻 Returning to pure camera motion

The simplest implementation became the strongest.

We introduced a completion state:

```
let completionPlayed = false;
```

Then in `draw()`:

```
if (!completionPlayed && discoveredCount() === discoverables.length) {  
  completionPlayed = true;  
}
```

And modified the HUD:

```
if (!completionPlayed) {  
  fill(COL_TEXT, 150);  
  text(`Discovered ${discoveredCount()}/${discoverables.length}`, 28, 92);  
}
```

AI:

No animation. No particles. No sound.

Just disappearance.

User:

When the counter vanished, it felt strangely calm.

AI:

That reaction confirms the design worked.

Instead of adding something, you removed something.

- 🎯 The system acknowledges completion
- 🧑🏻 The interface retreats
- 🌊 The camera continues breathing

User:

I realized that adding more would actually weaken the piece.

AI:

Exactly.

Throughout this project, the strongest decisions were acts of restraint:

- Choosing automatic motion over player control
- Choosing easing over randomness
- Choosing pause over constant motion
- Choosing drift over jitter
- Choosing subtle zoom over dramatic movement
- Choosing quiet disappearance over confetti

Each step avoided escalation.

User:

It doesn't feel like a "game" anymore. It feels like a small experience.

AI:

And that was the original intention.

By the end, the system contained:

- 🌍 A world larger than the screen
- 📽️ Authored, breathing camera motion
- 🌀 Organic drift
- 👁️ Quiet discoverables
- 🐢 Attention-based pacing shifts
- 🎬 Minimal cinematic framing
- 🧑 Reduced UI presence
- 🛑 A restrained completion state

No feature was added for spectacle.

Each was evaluated against tone.

User:

I think this is where it should stop.

AI:

And stopping at the right time is part of design maturity.

The final version wasn't the one with the most effects.

It was the one with the most coherence.